

Express.js

Introduction to Express.js

Express is developed by many developers around the world.

It is a **fast, minimalist web framework for Node.js**.

- Lightweight and flexible routing framework
 - Minimal core features
 - Extended using **Express middleware modules**
-

Installing Express

```
npm install express
```

OR

```
yarn add express
```

Advantage of Express over HTTP Server

HTTP Server

- Core built-in Node.js module
- Low-level control over requests and responses
- Manual handling of routing, headers, status codes, JSON parsing, etc.

Express Server

- Framework built on top of HTTP module
 - Simplifies routing, middleware, JSON handling
 - Cleaner, shorter, maintainable code
-

Routing in Express

Routing defines how an application responds to client requests using:

- URL path

- HTTP method

Route Syntax

```
app.METHOD(PATH, HANDLER)
```

Where:

1. **app** – Express instance
2. **METHOD** – HTTP method
3. **PATH** – Server path
4. **HANDLER** – Callback function

Basic Routing Example

```
const express = require("express");
const app = express();

app.get("/", (req, res) => {
  res.send("Welcome to Home Page");
});

app.get("/about", (req, res) => {
  res.send("About Us Page");
});

app.post("/login", (req, res) => {
  res.send("Login data received");
});

app.listen(3000, () => {
  console.log("Server running on port 3000");
});
```

express.Router()

Used to create modular route handlers.

user.js

```
const express = require("express");
const router = express.Router();

router.get("/", (req, res) => {
  res.send("User List");
});

router.get("/:id", (req, res) => {
  res.send(`User ID: ${req.params.id}`);
});

module.exports = router;
```

server.js

```
const express = require("express");
const app = express();
const userRouter = require("./user");

app.use("/users", userRouter);

app.listen(3000, () => {
  console.log("Server running on port 3000");
});
```

Middleware

Middleware runs between request and response.

Uses

- CORS handling
- JSON parsing (`express.json()`)
- Authentication
- Logging
- Database connectivity

MySQL Connection

Install

```
npm install mysql2
```

Create Pool

```
const mysql = require("mysql2");

const pool = mysql.createPool({
  host: "localhost",
  user: "root",
  password: "root",
  database: "kdac_db",
});
```

Query

```
const sql = "SELECT * FROM product";

pool.query(sql, (error, data) => {
  if (data) response.send(data);
  else response.send(error);
});
```
